AUTOMOBILE ENGINEERING · PART 8 OF 12

Software, Connectivity and Driver Assistance

By Isagi · 9 November 2026 · All specs as of November 2026 (manufacturer brochures and listings)

TL;DR

A modern car is a computer network on wheels. Dozens of small computers (ECUs) run the engine, brakes, airbags and screens, and talk to each other over an internal network. Newer cars add internet connectivity for apps, remote functions and over-the-air (OTA) updates, and cameras and radars for **ADAS** (advanced driver assistance systems). In India the ADAS you can buy today is Level 2 on the international scale: the car can help with steering and speed, but **you** remain responsible. ADAS has reached mainstream prices (the Mahindra XUV 3XO offered Level 2 ADAS from ₹11.99 lakh, in its AX5 L variant, at its April 2024 launch), but it was designed for orderly roads, and Indian roads are not always orderly.

1. The car as a network of computers

Each function has its own small computer, an **electronic control unit (ECU)**: engine management, transmission, ABS and ESC, airbags, air conditioning, instrument cluster, infotainment. A premium car can have well over a hundred.* They share data over an in-vehicle network, usually the **CAN bus**, so that for example the wheel-speed signal used by ABS is also available to the stability controller and the speedometer. The diagnostic port under the dashboard (OBD) lets a mechanic read fault codes from these ECUs, and BS6 norms made on-board diagnostics stricter (Part 2).

The industry buzzword is the **software-defined vehicle**: features that were once fixed in hardware are controlled by software that can be updated, and in some cases unlocked later through a paid subscription. That changes the business model: a car can earn revenue after the sale, and it also means two cars with identical hardware can differ in what they let you use.

2. Connectivity: from screens to over-the-air updates

- **Infotainment:** touchscreen, Android Auto and Apple CarPlay, voice assistants and navigation. Screen size and speed of response are now top reasons people choose one variant over another.
- **Connected-car apps:** a built-in SIM card links the car to a phone app for remote lock and unlock, location, climate pre-cooling and trip data. Most major brands offer one (Hyundai Bluelink, Kia Connect, MG iSMART, Tata iRA, Mahindra Adrenox, Maruti Suzuki Connect).*
- **OTA updates:** like a phone, a car can download new software and fix bugs without a visit to the workshop. Some updates are minor (maps, screen layouts); others change how the car drives or charges.

The trade-offs: a connected car collects data about where you go and how you drive. India's Digital Personal Data Protection Act, 2023 applies to personal data collected this way, although the rules for vehicles specifically are still evolving.* Anything online can in principle be attacked, so makers must secure the network. And long-term support matters: a car sold today must still receive updates in eight years.

3. ADAS: how far automation goes

Engineers describe how much of the driving a car does with a six-level scale from SAE International, from Level 0 (nothing) to Level 5 (everything, everywhere).

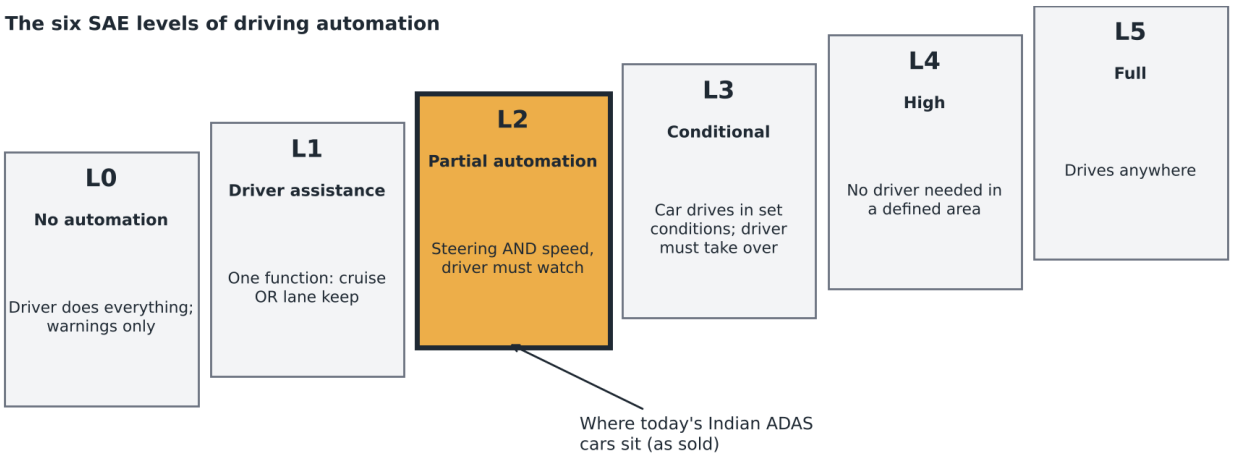


Fig 1. The six levels of driving automation defined by SAE International. Cars sold in India with ADAS are Level 2.

At **Level 2**, the car can steer and control speed together, for example adaptive cruise control with lane centring. But the driver must keep watching and be ready to take over at once. Levels 3 to 5 would let the car take responsibility in some or all conditions; they are not yet part of ordinary Indian car sales.*

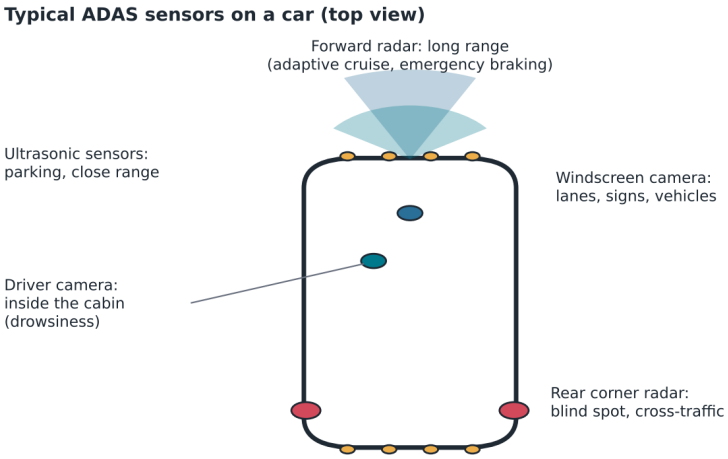


Fig 2. Where ADAS sensors typically sit. A forward camera and radar do most of the work; ultrasonic sensors handle parking; rear corner radars watch blind spots.

- **Autonomous emergency braking (AEB):** brakes if a collision is imminent. It is the most valuable feature, as Part 5 notes.
- **Adaptive cruise control (ACC):** keeps a set distance from the car in front, slowing and speeding up with traffic.
- **Lane departure warning and lane keep assist:** alert you or nudge the steering if you drift.
- **Blind-spot monitoring and rear cross-traffic alert:** warn about cars you cannot see.
- **Driver monitoring:** a cabin camera detects drowsiness or distraction.

4. ADAS meets Indian roads

ADAS was largely developed on roads with clear lane markings, predictable traffic and few unexpected obstacles. In India, markings are often faded or missing, two-wheelers filter through gaps, pedestrians and animals appear suddenly, and construction zones change the road layout. Lane-keeping depends on seeing markings, and emergency braking can trigger falsely or miss an unusual obstacle. Makers tune systems for local conditions, but owners often report that features behave inconsistently, so expect assistance, not autopilot.*

5. A buyer's checklist

- Check **which** ADAS functions are included and in which variant; the same name can cover very different feature sets.
- Test them in real traffic on a test drive, not only on an empty road.
- Ask how long software updates are supported and whether any features need a subscription.
- Read the data and privacy terms of the connected-car app.
- Treat ADAS as a safety net, not a replacement for attention.

Summary

- A car is a network of ECUs sharing data over a CAN bus; software increasingly defines what the car can do.
- Connectivity brings apps, remote features and OTA updates, and raises questions about data, security and long-term support.
- ADAS on sale in India is Level 2: the car assists, the driver stays responsible.
- AEB is the most valuable ADAS feature; others add comfort.
- Indian road conditions challenge systems designed for orderly roads; test them in real traffic.

*Unverified or general knowledge: the number of ECUs in premium cars, the list of connected-car brands, the status of data-protection rules for vehicles, the absence of Level 3+ cars in ordinary Indian sale and the statements on how ADAS behaves on Indian roads are general knowledge or reported owner experience, not tested here. Cross-checked in November 2026: the XUV 3XO launched on 29 April 2024 at an introductory ₹7.49 lakh for the base MX1 variant, which has no ADAS; the cheapest ADAS variant at launch was the AX5 L at ₹11.99 lakh (Autocar India, Mahindra press release); current range ₹7.79–15.04 lakh. SAE levels: SAE International standard J3016.

See also: the rest of the Automobile Engineering series (parts 1 to 12). Next: Part 9.